GNN Exercise: Neutrino Interaction Position Reconstruction

This is a quick summary of the final plots from the GNN model.

The model uses three DynamicEdgeConv layers, then global mean pooling, then a small MLP to predict the interaction point in the (x, y) plane.

The plotting script only saves residual plots for x, so that is all included here.

Training Curve

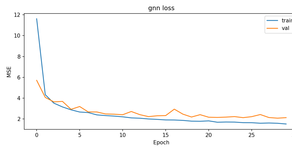


Figure 1: Training and validation MSE over epochs. Just a quick check that training behaves sensibly.

1. Overall Position Reconstruction

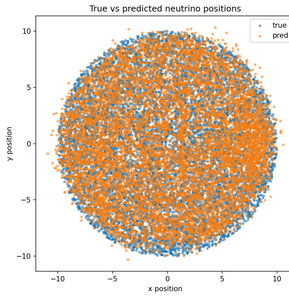
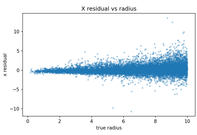
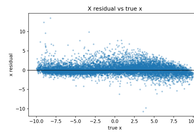


Figure 2: True and predicted positions in the (x, y) plane for the test set. This gives the overall picture.

2. Residual Diagnostics



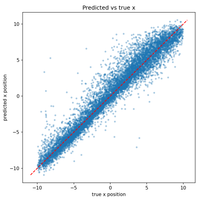
(a) x residual vs true radius.



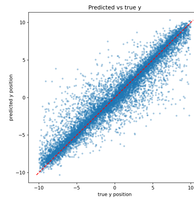
(b) x residual vs true x.

Figure 3: Two simple x residual checks. The left plot looks for radial bias, and the right one looks for any clear left–right shift.

3. Coordinate-wise Prediction Quality



(a) Predicted vs true x. The dashed red line is perfect prediction.



(b) Predicted vs true y. The dashed red line is perfect prediction.

Figure 4: Predicted versus true values for each coordinate. Closer to the diagonal is better.